

Juan Antonio Rodriguez Reyna

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EDUCATION

Tecnologico de Monterrey, Campus Monterrey (ITESM)

Monterrey, Nuevo Leon/Mexico

B.S. in Computer Science and Technology: 5th semester

Expected June 2028

- **Cumulative GPA:** 92 / 100
- **Certifications:** Bilingual IB Diploma, Snowflake SnowPro Core Certification (2025)
- **Relevant Coursework:** Business and Management HL, Data Science for Decision-Making, Object-Oriented Programming, Algorithms and Data Structures , Software/Web Development, Advanced Algorithms

WORK EXPERIENCE

ITESM – Data Engineer Specialist

Sept 2025 - Dec 2025

- Maintained and optimized **institutional data pipelines** processing **thousands of student records**, improving **data reliability** for academic and administrative reporting.
- Automated **data validation** and **transformation workflows** using **Python, R** and **SQL**, ensuring high data accuracy across multiple institutional datasets.
- Collaborated with **cross-functional teams** to translate complex analyses into **actionable insights**, supporting operational decision-making across the university.

PROJECTS

Oasis Smart Hydration System | ESP32, MQTT, SQL, Android, Gemini AI, HX711, BNO055

- **Designed the data architecture** for an **IoT hydration monitoring platform**, implementing the **SQL database** and backend logic that enabled real-time synchronization between embedded devices and the Android application.
- **Developed key software components** across the embedded system, mobile application, and AI-powered recommendation pipeline, contributing to a solution that achieved **≤5% measurement error**, **~90% drinking event detection accuracy**, and **210–300 ms** end-to-end latency.
- **Selected among 100+ engineering teams** to present at **Expo Ingenierías**, demonstrating an end-to-end system integrating **embedded systems, cloud infrastructure, and AI-driven personalization**.

EnSeñame | Swift, Vision Framework, Core ML, Gemini AI

- **Engineered** an AI-powered iOS application that teaches **Mexican Sign Language (LSM)** through **real-time hand gesture recognition** using **Vision Framework** and **Core ML**.
- **Designed** a gamified learning experience featuring interactive lessons, AI-assisted feedback, and a sign dictionary, transforming sign language practice into a **Duolingo-inspired** learning platform.

AWAQ | ASP.NET Core MVC, Unity, C#, SQL

- **Integrated** a **Unity WebGL** game with an **ASP.NET Core MVC** platform, enabling persistent player progress and seamless synchronization between gameplay and web services.
- **Architected** backend services and the **SQL data model** supporting authentication, progress tracking, and administrative management for the educational platform.

LEADERSHIP EXPERIENCE

SEITC – Computer Science Student Society, ITESM - President

May 2026 - Present

- **Elected President** of the **Computer Science Student Society (SEITC)**, leading an **8-member executive board** responsible for academic, professional, and community initiatives **serving 700+ Computer Science and prospective computing students** at Tecnológico de Monterrey, Campus Monterrey.
- **Directed** the organization's annual technical strategy by leading workshops, interview preparation programs, networking events, and industry collaborations designed to prepare students for careers in software engineering.
- **Designed** a multi-stage recruitment process, interviewed **20+ applicants** and restructured the organization by introducing new executive roles in **Education** and **Industry Relations**.
- **Launched** a web development outreach program teaching **HTML, CSS, and JavaScript** through hands-on workshops, expanding access to introductory computer science education for middle and high school students.

SKILLS

Programming Languages: Python, C++, C#, JavaScript, SQL, Swift, R, MATLAB

Frameworks & Technologies: ASP.NET Core MVC, FastAPI, React, Vision Framework, Core ML, MQTT, REST APIs

Databases: MySQL, Snowflake

Developer Tools: Git, GitHub, VS Code, Xcode, Unity, Arduino IDE

Languages: Spanish (Native), English (Fluent), Italian (Proficient)